

DeepPDF: Advanced Retrieval Strategies in RAG Systems

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Abstract

Retrieval-Augmented Generation (RAG) combines parametric memory with external non-parametric retrieval. Standard dense retrieval suffers from vocabulary mismatch, lost-in-the-middle phenomena, and low keyword fidelity. This paper introduces DeepPDF, a benchmarked framework comparing 7 advanced retrieval techniques.

1. Dense Vector Retrieval Baseline

Dense retrieval maps document chunks into a continuous embedding space using sentence transformers. Cosine similarity is computed: $\text{sim}(q, d) = (q \cdot d) / (\|q\| * \|d\|)$. Typical dimensionality is 384 dims.

2. BM25 Lexical Keyword Search

BM25 evaluates term frequency (TF) and inverse document frequency (IDF) with length normalization. Parameters $k_1=1.5$ and $b=0.75$ govern saturation and document length penalties.

3. Hybrid Search and Reciprocal Rank Fusion

Hybrid search merges sparse lexical retrieval (BM25) with dense vector retrieval.

Reciprocal Rank Fusion (RRF) computes: $RRF_score(d) = \sum (1 / (60 + rank_i(d)))$.

This achieves robust keyword matching (exact codes, acronyms) and deep semantic relevance.

4. Multi-Query Expansion & Sub-Query Decomposition

LLM generates 3-5 query variants from different perspectives to mitigate single-query bias.

Documents from all generated queries are fetched and deduplicated via union scoring.

5. Hypothetical Document Embeddings (HyDE)

HyDE instructs an LLM to generate a hypothetical answer document before embedding.

The hypothetical document is embedded, aligning the query closer to document passage vectors in latent space.

6. Two-Stage Cross-Encoder Re-Ranking

First stage retrieves a wide candidate pool (top-25) using fast bi-encoder similarity.

Second stage computes all-to-all cross-attention between question and passage: $\text{score} = \text{CrossEncoder}(q, d)$.

Top-k candidates (e.g. $k=5$) with highest re-ranking scores are provided to the generator.

7. Hierarchical Parent-Child Chunking

Documents are split into small 200-character child chunks for fine-grained vector similarity search.

Upon retrieval, child chunks map to their 1200-character parent chunk to give rich context to the LLM.

8. Experimental Benchmark Results

Cross-Encoder Re-ranking achieved highest MRR@5 (0.892) and NDCG@5 (0.914).

Hybrid RRF provided the best balance of latency (48ms) and Hit Rate@5 (96.4%).